

Chapter 9

Expected Value of a Discrete Random Variable

Lecture 9 | Devore 9e: 3.3

9.1 Leftovers on the cdf

Everything about X can be read off $F_X(x) = \mathbb{P}\{X \leq x\}$, provided you are careful about which endpoints are included. Write

$$F_X(x^-) := \lim_{y \rightarrow x^-} F_X(y)$$

for the left limit (the value “just before” any jump). Then

$$\begin{aligned}\mathbb{P}\{X > x\} &= 1 - F_X(x), \\ \mathbb{P}\{X < x\} &= F_X(x^-), \\ \mathbb{P}\{X = x\} &= F_X(x) - F_X(x^-), \\ \mathbb{P}\{a < X \leq b\} &= F_X(b) - F_X(a), \\ \mathbb{P}\{a \leq X \leq b\} &= F_X(b) - F_X(a^-).\end{aligned}$$

Derivation of the last one.

$$\{a \leq X \leq b\} = \{X \geq a\} \cap \{X \leq b\} = \{X \leq b\} \cap \{X < a\}^c = \{X \leq b\} \setminus \{X < a\},$$

and $\{X < a\} \subset \{X \leq b\}$, so the probabilities subtract: $\mathbb{P}\{a \leq X \leq b\} = \mathbb{P}\{X \leq b\} - \mathbb{P}\{X < a\} = F_X(b) - F_X(a^-)$. $\square$

Remark 9.1. For a discrete X the left limit is easy to evaluate: if X takes only the values $1, 2, 3, \dots$ then $\mathbb{P}\{X < 3\} = \mathbb{P}\{X \leq 2\} = F_X(2)$. If X can take the value 2.8 but nothing in $(2.8, 2.9]$, then $\mathbb{P}\{X < 2.9\} = \mathbb{P}\{X \leq 2.8\} = F_X(2.8)$. In words: step back to the largest value X can actually attain.

9.2 Expected value

Example 9.2 (Where the definition comes from). Among 100 students, the number registered for 0, 1, 2, 3, 4 courses is 5, 10, 30, 30, 25 respectively. On average, how many courses does a student

take?

$$\frac{\text{total courses}}{\text{total students}} = \frac{5 \cdot 0 + 10 \cdot 1 + 30 \cdot 2 + 30 \cdot 3 + 25 \cdot 4}{100} = 0.05 \cdot 0 + 0.10 \cdot 1 + 0.30 \cdot 2 + 0.30 \cdot 3 + 0.25 \cdot 4 = 2.6.$$

Look at the middle expression: dividing through by 100 turned the counts into *proportions*, i.e. into a pmf. So the average is a **probability-weighted sum of the values**.

Definition 9.3 (Expected value). Let X be a discrete random variable taking values in $D = \{x_1, x_2, \dots\}$ with $p_i = \mathbb{P}\{X = x_i\}$. The **expected value** (mean) of X is

$$\mathbb{E}[X] = \mu_X := \sum_{x_i \in D} x_i p_i = \sum_{x_i \in D} x_i \mathbb{P}\{X = x_i\},$$

provided the sum converges absolutely.

Remark 9.4. X is a *function*; $\mathbb{E}[X]$ is a *number*. Also, $\mathbb{E}[X]$ need not be a value X can take — no student registers for 2.6 courses.

Example 9.5 (Expectation generalizes the plain average). Given numbers $a_1 \leq a_2 \leq \dots \leq a_n$, define X by $\mathbb{P}\{X = a_i\} = \frac{1}{n}$. Then

$$\mathbb{E}[X] = \sum_{i=1}^n a_i \cdot \frac{1}{n} = \frac{a_1 + \dots + a_n}{n},$$

the ordinary mean. The general definition just allows unequal weights.

9.2.1 A worked series: the geometric mean

Example 9.6 (Shooting at a target). $\mathbb{P}\{X = n\} = (1 - \beta)^{n-1} \beta$ for $n \geq 1$. Then

$$\mathbb{E}[X] = \sum_{n=1}^{\infty} n(1 - \beta)^{n-1} \beta = \beta \sum_{n=1}^{\infty} n(1 - \beta)^{n-1}.$$

The trick is to recognize $n(1 - \beta)^{n-1} = -\frac{d}{d\beta}(1 - \beta)^n$:

$$\mathbb{E}[X] = -\beta \frac{d}{d\beta} \sum_{n=1}^{\infty} (1 - \beta)^n = -\beta \frac{d}{d\beta} \left(\frac{1 - \beta}{1 - (1 - \beta)} \right) = -\beta \frac{d}{d\beta} \left(\frac{1 - \beta}{\beta} \right) = -\beta \frac{d}{d\beta} \left(\frac{1}{\beta} - 1 \right) = -\beta \left(-\frac{1}{\beta^2} \right) = \boxed{\frac{1}{\beta}}.$$

(We used $\sum_{n \geq 1} a^n = \frac{a}{1-a}$ for $|a| < 1$, and differentiated term by term, which is legitimate inside the radius of convergence.)

Sanity check: if you hit 10% of the time you expect to need 10 shots. Good.

9.2.2 When the mean does not exist

Example 9.7 (Heavy tail). Let X be the number of interviews needed to get a job, with

$$\mathbb{P}\{X = x\} = a \cdot \frac{1}{x^2}, \quad x = 1, 2, 3, \dots$$

First find a . We need $\sum_{x \geq 1} a/x^2 = 1$, and $\sum_{x \geq 1} x^{-2} = \pi^2/6$, so $a = 6/\pi^2$. That is a perfectly legitimate pmf. But

$$\mathbb{E}[X] = \sum_{x=1}^{\infty} x \cdot \frac{6}{\pi^2} \frac{1}{x^2} = \frac{6}{\pi^2} \sum_{x=1}^{\infty} \frac{1}{x} = \infty.$$

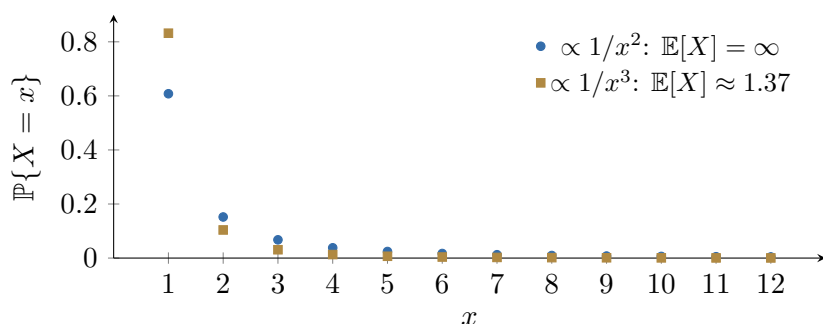
Remark 9.8. The mean is $+\infty$: the distribution has a **heavy tail**. Large values are individually unlikely but not unlikely *enough* to keep the weighted sum finite. This is not a pathology invented by mathematicians — insurance losses, file sizes, city populations, and income all behave this way, and for such data the sample mean is a badly behaved summary.

Example 9.9 (How much lighter must the tail be?). Keep the same shape but decay one power faster: let

$$\mathbb{P}\{X = x\} = \frac{b}{x^3}, \quad x = 1, 2, 3, \dots$$

Normalizing needs $\sum_{x \geq 1} x^{-3} = \zeta(3) \approx 1.2021$, so $b = 1/\zeta(3) \approx 0.832$. Now

$$\mathbb{E}[X] = \sum_{x=1}^{\infty} x \cdot \frac{b}{x^3} = b \sum_{x=1}^{\infty} \frac{1}{x^2} = b \cdot \frac{\pi^2}{6} \approx 0.832 \times 1.645 \approx 1.37 < \infty.$$



Remark 9.10. Both pmfs look like they decay to zero, and on a plot they are nearly indistinguishable past $x = 5$. Yet one has an infinite mean and the other does not. “Heavy tail” is therefore not something you can reliably eyeball — it is a statement about how fast $\sum x p(x)$ accumulates out in the tail, and the boundary sits exactly at the $1/x^2$ rate: for $p(x) \propto x^{-s}$ the sum $\sum x p(x)$ behaves like $\sum x^{1-s}$, which converges precisely when $s > 2$.

9.3 Functions of a random variable

If $X : S \rightarrow \mathbb{R}$ is a random variable and $h : \mathbb{R} \rightarrow \mathbb{R}$, then the composition $h(X) = h \circ X : S \rightarrow \mathbb{R}$ is again a random variable. Its expectation can be computed without ever finding the distribution of $h(X)$:

Theorem 9.11 (Law of the unconscious statistician). *If X takes values in $D = \{x_1, x_2, \dots\}$ and $h : \mathbb{R} \rightarrow \mathbb{R}$, then*

$$\mathbb{E}[h(X)] = \sum_{x_i \in D} h(x_i) \mathbb{P}\{X = x_i\}.$$

Example 9.12. Shooting at a target, X = number of shots. Each bullet costs 5, and maintenance costs $\log X$. Total expense is $h(X) = 5X + \log X$:

X	1	2	3	...
$h(X)$	$5 + \log 1$	$10 + \log 2$	$15 + \log 3$	...
$\mathbb{P}$	β	$(1 - \beta)\beta$	$(1 - \beta)^2\beta$	...

so $\mathbb{E}[h(X)] = \sum_{n \geq 1} (5n + \log n)(1 - \beta)^{n-1}\beta$. Note we did *not* have to work out the pmf of $h(X)$ — that is exactly what the theorem buys us.

Corollary 9.13 (Linearity in the simplest case). *For constants $a, b \in \mathbb{R}$,*

$$\mathbb{E}[aX + b] = a \mathbb{E}[X] + b.$$

Proof.

$$\mathbb{E}[aX + b] = \sum_{x_i \in D} (ax_i + b) \mathbb{P}\{X = x_i\} = a \underbrace{\sum_{x_i \in D} x_i \mathbb{P}\{X = x_i\}}_{= \mathbb{E}[X]} + b \underbrace{\sum_{x_i \in D} \mathbb{P}\{X = x_i\}}_{= 1} = a \mathbb{E}[X] + b. \quad \square$$

Remark 9.14. Linearity is what makes expectation usable. Note carefully that $\mathbb{E}[h(X)] \neq h(\mathbb{E}[X])$ in general — for instance $\mathbb{E}[X^2] \neq (\mathbb{E}[X])^2$, and the gap between them is exactly the variance, which is next lecture.

9.4 Sums of random variables

The corollary handles one random variable. The real power of linearity is that it extends to *sums of several* random variables, with no assumptions at all.

When the sample space is finite or countably infinite, an expectation can be computed outcome by outcome. Grouping the outcomes by the value X gives them,

$$\mathbb{E}[X] = \sum_x x \mathbb{P}\{X = x\} = \sum_x x \sum_{\omega: X(\omega)=x} \mathbb{P}(\{\omega\}) = \sum_{\omega \in S} X(\omega) \mathbb{P}(\{\omega\}),$$

the regrouping being legitimate whenever the sum converges absolutely.

Theorem 9.15 (Linearity of expectation). *For random variables $X_1, \dots, X_n$ on the same probability space, each with a finite expectation,*

$$\mathbb{E}[X_1 + X_2 + \dots + X_n] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \dots + \mathbb{E}[X_n],$$

whether or not the X_i are independent.

Proof for a finite or countable S . $\mathbb{E}[X + Y] = \sum_{\omega} (X(\omega) + Y(\omega)) \mathbb{P}(\{\omega\}) = \mathbb{E}[X] + \mathbb{E}[Y]$, and induct on n . Lecture 20 states the result in general, continuous variables included. $\square$

Combined with indicators (Lecture 8), this becomes a machine for expected counts. To find the expected *number* of the events $A_1, \dots, A_n$ that happen, write the count as $N = \mathbf{1}_{A_1} + \dots + \mathbf{1}_{A_n}$. Since $\mathbb{E}[\mathbf{1}_A] = 1 \cdot \mathbb{P}(A) + 0 \cdot \mathbb{P}(A^c) = \mathbb{P}(A)$,

$$\mathbb{E}[N] = \mathbb{P}(A_1) + \dots + \mathbb{P}(A_n)$$

however dependent the events may be; the pmf of N is never needed. This **indicator method** is one of the most useful tricks in probability.

Example 9.16 (The hat check, again). In the hat check of Lecture 4, how many people get their own hat back, on average? With A_i the event that person i does, $\mathbb{P}(A_i) = \frac{1}{n}$, so $\mathbb{E}[N] = n \cdot \frac{1}{n} = 1$ — for every n . The events are far from independent (if the first $n - 1$ people all get their own hats, so does the last), and linearity does not care. Compare Lecture 4, where the probability of *no* match needed the whole inclusion–exclusion formula.

Example 9.17 (Runs in a shuffled deck). A deck of 26 red and 26 black cards is shuffled. A *run* is a maximal block of consecutive cards of one color; RRBRRRB has four runs. A new run begins at every color change, so

$$\text{runs} = 1 + \sum_{i=1}^{51} \mathbf{1}_{C_i}, \quad C_i = \{\text{cards } i \text{ and } i + 1 \text{ differ in color}\}.$$

Any two positions hold a uniformly random pair of distinct cards, so $\mathbb{P}(C_i) = 2 \cdot \frac{26}{52} \cdot \frac{26}{51} = \frac{26}{51}$, and

$$\mathbb{E}[\text{runs}] = 1 + 51 \cdot \frac{26}{51} = \boxed{27}.$$

Example 9.18 (Meeting in a tournament). 64 players enter a single-elimination tournament. In each round the remaining players are paired uniformly at random, and every match is won by either player with probability $\frac{1}{2}$. What is the probability that two particular players, Alice and Bob, play each other at some point?

Count matches instead of following Alice and Bob. Every match eliminates exactly one player and 63 must be eliminated, so *exactly 63 matches are played*, whatever happens. For each of the $\binom{64}{2} = 2016$ pairs of players, let I_{pair} indicate that the pair meets. Then $\sum_{\text{pairs}} I_{\text{pair}} = 63$ with certainty, and taking expectations,

$$\sum_{\text{pairs}} \mathbb{P}\{\text{pair meets}\} = 63.$$

The rules treat all players alike, so every pair meets with the same probability p . Hence $2016p = 63$ and $p = \boxed{\frac{1}{32}}$. (With 2^k players the same argument gives $2/2^k$.)

Remark 9.19. Following Alice and Bob round by round is painful: both must survive until the random draw finally pairs them. The indicator argument sidesteps all of that. Whenever a question asks for an *expected* count — or for a probability that can be read off from one, as here — look for indicators before you look for a pmf.